Supplementary Data Online

Artificial intelligence-based quantification of breast arterial calcifications to predict cardiovascular morbidity and mortality

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Supplementary Table 1 Patient Characteristics of the independent 500 exams for AI-model validation

	Value
Age, years, median (IQR)	60.4 (50.3 – 68.3)
Race, n (%)	
Asian	24 (4.82)
Black	256 (51.41)
Other	5 (0.6)
Unknown	24 (4.82)
White	191 (38.35)
Breast tissue density (BI-RADS), (%)	
A	14.66
B	41.57
C	38.15
D	5.62
Scanner Distribution, n (%)	
Hologic, Inc.	483 (96.6)
GE Medical Systems	17 (3.4)

Supplementary Table 2 ICD Codes for the MACE criteria

Outcome	ICD-9	ICD-10
Non-fatal myocardial infarction	410.0, 410.00, 410.01, 410.1, 410.10, 410.11, 410.20, 410.2, 410.21, 410.3, 410.30, 410.31, 410.40, 410.4, 410.41, 410.5, 410.50, 410.51, 410.6, 410.60, 410.61, 410.7, 410.70, 410.71, 410.8, 410.80, 410.81, 410.9, 410.90, 410.91	I21.0, I21.01, I21.02, I21.09, I21.1, I21.11, I21.19, I21.2, I21.21, I21.29, I21.3, I21.4, I21.9
Non-fatal ischemic stroke	433.01, 433.11, 433.21, 433.31, 433.81, 433.91, 434.01, 434.11, 434.91	I63.*
Acute Heart Failure	428.43, 428.33, 428.23, 428.41	I50.813, I50.43, I50.33, I50.23, I50.21

Pre-selected ICD codes extracted from EHR[1]
- Revascularization procedures may be influenced by regional practice variation, access to care, or incidental findings on imaging (e.g., CT or catheterization performed for non-acute reasons), rather than symptomatic or emergent cardiovascular events. Including revascularization might inflate event rates and obscure the association between BAC and spontaneous cardiovascular events. Therefore we elected to exclude these from endpoints.

Supplementary Table 3 Inclusion Criteria for Diabetes

We defined patients who have diabetes at their index mammogram using ICD codes, lab values or medication.

Criteria	Description
ICD-9	249.xx, 250.xx
ICD-10	E08.xx, E09.xx, E10.xx, E11.xx, E13.xx
Lab Values	Fasting Glucose ≥126mg/mL; Glucose ≥ 200mg/mL; A1C≥6.5%
Medications	Metformin, Glimepiride, Glipizide, Glyburide, Pioglitazone, Rosiglitazone, Acarbose, Miglitol, Repaglinide, Nateglinide, Sitagliptin, Saxagliptin, Insulin Lispro, Insulin Glulisine, NPH Insulin, Regular Insulin, Insulin Glargine, Insulin Detemir, Exenatide, Liraglutide, Pramlintide

Supplementary Table 4 Incidence Rates of Major Adverse Cardiovascular Events (MACE) per 1000 Person-Years.

Rates are presented for the full Emory and Mayo Clinic cohorts, stratified by Breast Arterial Calcification (BAC) severity. A separate analysis shows rates for the subgroup with available PREVENT scores, stratified by both BAC severity and PREVENT risk categories.

		Any		AMI		Stroke		HF		Death	
		Emory	**Mayo**	**Emory**	**Mayo**	**Emory**	**Mayo**	**Emory**	**Mayo**	**Emory**	**Mayo**
Across entire cohort (Emory n=74,124; Mayo n=49,638)											
BAC Severity	Zero (0 mm^2)	5.96	17.45	0.95	3.05	2.86	3.93	1.34	9.41	1.79	4.74
	Mild (>0-10 mm^2)	9.58	25.45	1.54	4.56	4.16	5.67	2.52	14.51	3.28	7.30
	Moderate (>10-25 mm^2)	24.98	58.39	3.82	9.10	11.20	9.30	6.37	36.53	10.50	18.21
	Severe (>25 mm^2)	48.89	97.13	6.65	17.00	19.68	19.03	20.18	58.55	17.83	38.03
Subgroup of patients who have PREVENT scores (Emory n=4,876; Mayo n=13,564)											
BAC severity	Zero (0 mm^2)	9.86	35.95	1.69	6.49	5.00	7.87	2.59	21.41	2.81	9.04
	Mild (>0-10 mm^2)	17.66	55.03	4.06	9.27	8.68	11.29	5.19	32.70	4.95	15.12
	Moderate (>10-25 mm^2)	31.19	90.54	4.72	13.80	15.69	16.20	13.65	52.63	10.12	29.54
	Severe (>25 mm^2)	66.09	154.35	7.58	27.32	20.67	26.87	26.67	85.82	31.18	57.87
PREVENT Risk Category	Low (<5 %)	3.22	27.52	0.73	5.33	1.63	7.24	0.67	14.98	0.78	7.08
	Borderline (≥5-7.5 %)	5.74	26.87	1.16	5.24	3.01	4.09	3.48	15.65	1.31	5.97
	Intermediate (≥7.5-20 %)	15.00	43.94	2.25	8.09	7.88	8.51	1.49	26.79	4.54	10.92
	High (≥20 %)	36.36	92.09	7.11	13.36	15.72	16.45	13.41	57.07	10.61	26.50

Supplementary Table 5 Subdistribution Hazards for Internal Cohort (Emory)

Model		Covariate	Composite (AMI+Stroke+HF)			AMI			Stroke			HF		
			N (%)	sHR (95% CI)	p-value	N (%)	sHR (95% CI)	p-value	N (%)	sHR (95% CI)	p-value	N (%)	sHR (95% CI)	p-value
CATEGORICAL														
BAC + Age	BAC	Zero (N=62,187)	1,622 (2.61)	Reference		422 (0.68)	Reference		1258 (2.02)	Reference		595 (0.96)	Reference	
		Mild (N=10,369)	393 (3.79)	1.154 (0.993 - 1.341)	0.061	112 (1.08)	1.010 (0.735 - 1.387)	0.952	300 (2.89)	1.191 (1.006 - 1.408)	0.042	183 (1.76)	1.458 (1.150 - 1.849)	0.002
		Moderate (N=1,097)	103 (9.39)	3.704 (2.468 - 5.559)	<0.001	29 (2.64)	4.031 (1.918 - 8.475)	<0.001	83 (7.57)	4.158 (2.717 - 6.363)	<0.001	48 (4.38)	6.852 (3.896 - 12.051)	<0.001
		Severe (N=471)	72 (15.29)	8.454 (5.813 - 12.296)	<0.001	20 (4.25)	10.265 (5.612 - 18.774)	<0.001	57 (12.10)	7.983 (5.179 - 12.304)	<0.001	58 (12.31)	19.022 (12.706 - 28.477)	<0.001
	Age	per 1 year increase		1.063 (1.058 - 1.068)	<0.001		1.059 (1.049 - 1.069)	<0.001		1.064 (1.058 - 1.070)	<0.001		1.079 (1.070 - 1.088)	<0.001
	Interaction	Age × Mild BAC		0.997 (0.986 - 1.008)	0.563		1.019 (0.997 - 1.042)	0.096		0.991 (0.979 - 1.004)	0.176		0.992 (0.975 - 1.008)	0.322
		Age × Moderate BAC		0.952 (0.928 - 0.977)	<0.001		0.952 (0.908 - 0.999)	0.044		0.946 (0.921 - 0.972)	<0.001		0.918 (0.883 - 0.954)	<0.001
		Age × Severe BAC		0.936 (0.913 - 0.959)	<0.001		0.918 (0.880 - 0.959)	<0.001		0.938 (0.912 - 0.965)	<0.001		0.926 (0.902 - 0.950)	<0.001
BAC + PREVENT	BAC	Zero (N=4,008)	255 (6.36)	Reference		68 (1.70)	Reference		198 (4.94)	Reference		104 (2.59)	Reference	
		Mild (N=711)	78 (10.97)	1.363 (1.051 - 1.767)	0.019	28 (3.94)	1.756 (1.121 - 2.752)	0.014	59 (8.30)	1.307 (0.972 - 1.758)	0.077	36 (5.06)	1.376 (0.933 - 2.029)	0.107
		Moderate (112)	19 (16.96)	1.576 (0.990 - 2.508)	0.055	5 (4.46)	1.372 (0.559 - 3.365)	0.490	16 (14.29)	1.677 (1.006 - 2.796)	0.047	14 (12.50)	2.316 (1.273 - 4.212)	0.006
		Severe (N=45)	11 (24.44)	2.125 (1.160 - 3.891)	0.015	3 (6.67)	1.819 (0.537 - 6.159)	0.336	8 (17.78)	1.917 (0.957 - 3.842)	0.067	10 (22.22)	3.890 (2.048 - 7.389)	<0.001
	PREVENT	Low (N=1,761)	41 (2.33)	Reference		13 (0.74)	Reference		29 (1.65)	Reference		12 (0.68)	Reference	
		Borderline (N=600)	24 (4.00)	1.715 (1.035 - 2.843)	0.036	7 (1.17)	1.547 (0.618 - 3.874)	0.351	18 (3.00)	1.809 (1.004 - 3.261)	0.048	9 (1.50)	2.169 (0.914 - 5.148)	0.079
		Intermediate (N=1,878)	179 (9.53)	4.078 (2.905 - 5.724)	<0.001	42 (2.24)	2.863 (1.547 - 5.300)	0.001	144 (7.67)	4.582 (3.072 - 6.833)	<0.001	65 (3.46)	4.771 (2.582 - 8.816)	<0.001
		High (N=637)	119 (18.68)	8.151 (5.678 - 11.701)	<0.001	42 (6.59)	8.074 (4.300 - 15.162)	<0.001	90 (14.13)	8.314 (5.430 - 12.730)	<0.001	78 (12.24)	15.841 (8.502 - 29.514)	<0.001
CONTINUOUS														
BAC + Age	BAC	per 1mm2 increase		1.093 (1.059 - 1.128)	<0.001		1.101 (1.054 - 1.151)	<0.001		1.091 (1.054 - 1.129)	<0.001		1.118 (1.091 - 1.147)	<0.001
	Age	per 1 year increase		1.062 (1.058 - 1.067)	<0.001		1.062 (1.054 - 1.071)	<0.001		1.062 (1.057 - 1.067)	<0.001		1.077 (1.070 - 1.085)	<0.001
	Interaction	Age × BAC		0.999 (0.999 - 1.000)	<0.001		0.999 (0.998 - 1.000)	0.003		0.999 (0.999 - 1.000)	0.001		0.999 (0.998 - 0.999)	<0.001
BAC + PREVENT	BAC	BAC (per 1mm2 increase)		1.017 (1.003 - 1.030)	0.013		1.015 (0.994 - 1.037)	0.171		1.017 (1.002 - 1.031)	0.026		1.030 (1.015 - 1.046)	<0.001
	PREVENT	PREVENT (per 1 point increase)		1.053 (1.046 - 1.060)	<0.001		1.058 (1.049 - 1.068)	<0.001		1.050 (1.042 - 1.058)	<0.001		1.063 (1.054 - 1.072)	<0.001
NON-LINEAR														
BAC + Age	log2(BAC+1)	Doubling of BAC		2.449 (1.874 - 3.201)	<0.001		2.443 (1.444 - 4.133)	0.001		2.602 (1.944 - 3.481)	<0.001		3.901 (2.805 - 5.426)	<0.001
	Age	per 1 year increase		1.063 (1.058 - 1.068)	<0.001		1.062 (1.053 - 1.072)	<0.001		1.063 (1.057 - 1.068)	<0.001		1.078 (1.070 - 1.087)	<0.001
	Interaction	Age × log2(BAC+1)		0.990 (0.986 - 0.993)	<0.001		0.990 (0.982 - 0.997)	0.008		0.989 (0.984 - 0.993)	<0.001		0.984 (0.979 - 0.989)	<0.001
BAC + PREVENT	log2(BAC+1)	Doubling of BAC		1.147 (1.059 - 1.242)	0.001		1.148 (1.006 - 1.311)	0.040		1.146 (1.049 - 1.253)	0.003		1.265 (1.140 - 1.404)	<0.001
	PREVENT	PREVENT (per 1 point increase)		1.051 (1.044 - 1.058)	<0.001		1.057 (1.047 - 1.067)	<0.001		1.049 (1.040 - 1.057)	<0.001		1.061 (1.052 - 1.071)	<0.001

Supplementary Table 6 Subdistribution Hazards for External Cohort (Mayo Clinic)

Model		Covariate	Composite (AMI+Stroke+HF)			AMI			Stroke			HF		
			N (%)	sHR (95% CI)	p-value	N (%)	sHR (95% CI)	p-value	N (%)	sHR (95% CI)	p-value	N (%)	sHR (95% CI)	p-value
CATEGORICAL														
BAC + Age	BAC	Zero (N=36,727)	3085 (8.39)	Reference		683 (1.86)	Reference		877 (2.39)	Reference		2065 (5.62)	Reference	
		Mild (N=10,725)	1289 (12.01)	1.046 (0.956 - 1.145)	0.330	297 (2.77)	1.074 (0.897 - 1.287)	0.430	367 (3.42)	1.037 (0.877 - 1.226)	0.670	912 (8.50)	1.096 (0.980 - 1.227)	0.110
		Moderate (N=1,582)	394 (24.9)	2.445 (1.951 - 3.064)	<0.001	86 (5.44)	2.295 (1.508 - 3.493)	<0.001	87 (5.50)	1.395 (0.837 - 2.325)	0.200	315 (19.91)	3.191 (2.488 - 4.092)	<0.001
		Severe (N=604)	214 (35.43)	4.151 (2.923 - 5.894)	<0.001	60 (9.93)	4.180 (2.203 - 7.932)	<0.001	66 (10.93)	6.330 (3.996 - 10.028)	<0.001	175 (28.97)	4.650 (3.070 - 7.041)	<0.001
	Age	per 1 year increase		1.079 (1.075 - 1.084)	<0.001		1.061 (1.052 - 1.070)	<0.001		1.066 (1.058 - 1.074)	<0.001		1.092 (1.086 - 1.098)	<0.001
	Interaction	Age × Mild BAC		1.006 (0.998 - 1.014)	0.120		1.012 (0.997 - 1.028)	0.120		1.010 (0.996 - 1.025)	0.160		1.001 (0.992 - 1.011)	0.820
		Age × Moderate BAC		0.964 (0.947 - 0.980)	<0.001		0.972 (0.942 - 1.002)	0.069		0.989 (0.953 - 1.027)	0.570		0.949 (0.931 - 0.967)	<0.001
		Age × Severe BAC		0.955 (0.931 - 0.979)	<0.001		0.971 (0.927 - 1.017)	0.210		0.923 (0.891 - 0.957)	<0.001		0.953 (0.925 - 0.982)	0.002
BAC + PREVENT	BAC	Zero (N=11,196)	1896 (16.91)	Reference		438 (3.91)	Reference		528 (4.72)	Reference		1372 (12.25)	Reference	
		Mild (N=1,910)	463 (24.24)	1.295 (1.169 - 1.434)	<0.001	106 (5.55)	1.267 (1.023 - 1.569)	0.030	128 (6.70)	1.293 (1.066 - 1.569)	0.009	344 (18.01)	1.281 (1.137 - 1.444)	<0.001
		Moderate (N=483)	167 (34.57)	1.752 (1.488 - 2.064)	<0.001	39 (8.07)	1.714 (1.226 - 2.395)	0.002	45 (9.32)	1.659 (1.210 - 2.274)	0.002	130 (26.92)	1.755 (1.457 - 2.113)	<0.001
		Severe (N=248)	116 (46.77)	2.601 (2.120 - 3.191)	<0.001	38 (15.32)	3.276 (2.315 - 4.637)	<0.001	37 (14.92)	2.655 (1.874 - 3.761)	<0.001	95 (38.31)	2.722 (2.168 - 3.419)	<0.001
	PREVENT	Low (N=6,616)	875 (13.22)	Reference		213 (3.22)	Reference		287 (4.34)	Reference		580 (8.77)	Reference	
		Borderline (N=885)	850 (96.04)	0.962 (0.792 - 1.169)	0.700	28 (3.16)	0.990 (0.667 - 1.468)	0.960	22 (2.49)	0.571 (0.370 - 0.881)	0.011	81 (9.15)	1.048 (0.831 - 1.321)	0.690
		Intermediate (N=3,969)	804 (20.25)	1.533 (1.393 - 1.687)	<0.001	193 (4.86)	1.449 (1.192 - 1.763)	<0.001	202 (5.09)	1.129 (0.943 - 1.353)	0.190	598 (15.07)	1.711 (1.526 - 1.919)	<0.001
		High (N=2,367)	113 (4.77)	2.836 (2.571 - 3.128)	<0.001	187 (7.90)	2.175 (1.769 - 2.673)	<0.001	227 (9.59)	2.016 (1.679 - 2.420)	<0.001	682 (28.81)	3.336 (2.973 - 3.743)	<0.001
CONTINUOUS														
BAC + Age	BAC	per 1mm2 increase		1.004 (1.003 - 1.004)	<0.001		1.004 (1.003 - 1.005)	<0.001		1.004 (1.003 - 1.005)	<0.001		1.004 (1.003 - 1.005)	<0.001
	Age	per 1 year increase		1.081 (1.077 - 1.085)	<0.001		1.064 (1.057 - 1.072)	<0.001		1.069 (1.062 - 1.076)	<0.001		1.092 (1.088 - 1.097)	<0.001
	Interaction	Age × BAC		1.000 (1.000 - 1.000)	<0.001		1.000 (1.000 - 1.000)	0.046		1.000 (1.000 - 1.000)	0.001		1.000 (1.000 - 1.000)	<0.001
BAC + PREVENT	BAC	BAC (per 1mm2 increase)		1.003 (1.002 - 1.004)	<0.001		1.004 (1.003 - 1.005)	<0.001		1.004 (1.003 - 1.005)	<0.001		1.003 (1.002 - 1.004)	<0.001
	PREVENT	PREVENT (per 1 point increase)		1.039 (1.036 - 1.043)	<0.001		1.031 (1.024 - 1.037)	<0.001		1.025 (1.018 - 1.031)	<0.001		1.046 (1.042 - 1.050)	<0.001
NON-LINEAR														
BAC + Age	log_2(BAC+1)	Doubling of BAC		1.088 (1.067 - 1.109)	<0.001		1.092 (1.050 - 1.135)	<0.001		1.077 (1.038 - 1.117)	<0.001		1.116 (1.090 - 1.143)	<0.001
	Age	per 1 year increase		1.080 (1.076 - 1.085)	<0.001		1.062 (1.053 - 1.071)	<0.001		1.068 (1.060 - 1.076)	<0.001		1.093 (1.087 - 1.098)	<0.001
	Interaction	Age × log2(BAC+1)		0.998 (0.996 - 0.999)	0.007		0.999 (0.996 - 1.002)	0.570		0.998 (0.995 - 1.001)	0.190		0.997 (0.995 - 0.998)	<0.001
BAC + PREVENT	log_2(BAC+1)	Doubling of BAC		1.113 (1.093 - 1.133)	<0.001		1.108 (1.066 - 1.152)	<0.001		1.126 (1.088 - 1.166)	<0.001		1.121 (1.098 - 1.145)	<0.001
	PREVENT	PREVENT (per 1 point increase)		1.038 (1.034 - 1.041)	<0.001		1.030 (1.023 - 1.037)	<0.001		1.023 (1.017 - 1.030)	<0.001		1.044 (1.040 - 1.048)	<0.001

Supplementary Table 7 Non-linear analysis of BAC and clinical outcomes

			ANY		AMI		Stroke		HF		Death	
INTERNAL COHORT (Emory)			**HR**	**p-value**	**HR**	**p-value**	**HR**	**p-value**	**HR**	**p-value**	**HR**	**p-value**
BAC + Age	log_2(BAC+1)	Doubling of BAC	1.131 (1.097 - 1.166)	<0.001	1.068 (1.030 - 1.107)	<0.001	1.102 (1.067 - 1.139)	<0.001	1.129 (1.091 - 1.168)	<0.001	1.134 (1.096 - 1.172)	<0.001
	Age	per 1 year increase	1.027 (1.025 - 1.030)	<0.001	1.008 (1.005 - 1.011)	<0.001	1.018 (1.015 - 1.020)	<0.001	1.013 (1.010 - 1.016)	<0.001	1.013 (1.010 - 1.015)	<0.001
	Age × log_2(BAC+1)		1.002 (1.001 - 1.002)	<0.001	1.001 (1.000 - 1.002)	<0.001	1.001 (1.001 - 1.002)	<0.001	1.002 (1.001 - 1.002)	<0.001	1.002 (1.001 - 1.002)	<0.001
BAC + PREVENT	log_2(BAC+1)	Doubling of BAC	1.173 (1.108 - 1.241)	<0.001	1.113 (1.007 - 1.230)	0.036	1.138 (1.056 - 1.226)	0.001	1.219 (1.122 - 1.324)	<0.001	1.204 (1.104 - 1.313)	<0.001
	PREVENT	per 1 point increase	1.049 (1.043 - 1.055)	<0.001	1.030 (1.019 - 1.041)	<0.001	1.037 (1.029 - 1.045)	<0.001	1.043 (1.034 - 1.053)	<0.001	1.039 (1.030 - 1.049)	<0.001
EXTERNAL COHORT (Mayo Clinic)												
BAC + Age	log_2(BAC+1)	Doubling of BAC	1.089 (1.068 - 1.111)	<0.001	1.065 (1.041 - 1.090)	<0.001	1.059 (1.035 - 1.083)	<0.001	1.098 (1.076 - 1.121)	<0.001	1.103 (1.080 - 1.128)	<0.001
	Age	per 1 year increase	1.036 (1.034 - 1.038)	<0.001	1.010 (1.008 - 1.013)	<0.001	1.013 (1.010 - 1.015)	<0.001	1.029 (1.027 - 1.031)	<0.001	1.017 (1.014 - 1.019)	<0.001
	Age × log_2(BAC+1)		1.001 (1.001 - 1.002)	<0.001	1.001 (1.001 - 1.001)	<0.001	1.001 (1.001 - 1.001)	<0.001	1.002 (1.001 - 1.002)	<0.001	1.001 (1.001 - 1.002)	<0.001
BAC + PREVENT	log_2(BAC+1)	Doubling of BAC	1.132 (1.108 - 1.156)	<0.001	1.084 (1.050 - 1.119)	<0.001	1.085 (1.052 - 1.120)	<0.001	1.132 (1.106 - 1.159)	<0.001	1.170 (1.137 - 1.203)	<0.001
	PREVENT	per 1 point increase	1.027 (1.024 - 1.030)	<0.001	1.011 (1.007 - 1.015)	<0.001	1.010 (1.006 - 1.014)	<0.001	1.029 (1.026 - 1.032)	<0.001	1.022 (1.018 - 1.025)	<0.001

Supplementary Table 8 Characteristics of Patients with PREVENT Scores by MACE Outcome

		INTERNAL COHORT (Emory)			EXTERNAL COHORT (Mayo Clinic)		
		MACE (-)	MACE (+)	P-Value	MACE (-)	MACE (+)	P-Value
		N=4,393 (90.1%)	N=483 (9.9%)		N=9,995 (73.7%)	N=3,569 (26.3%)	
Age, mean (SD)		59.1 (10.0)	65.0 (9.2)	<0.001	64.1 (10.0)	69.8 (9.8)	<0.001
Race, n (%)	**Asian**	122 (2.8)	3 (0.6)	<0.001	169 (1.7)	41 (1.1)	<0.001
	Black	2,252 (51.3)	316 (65.4)		165 (1.7)	39 (1.1)	
	Other	30 (0.7)	0 (0.0)		145 (1.5)	43 (1.2)	
	Unknown	83 (1.9)	1 (0.2)		15 (0.2)	1 (0.0)	
	White	1906 (43.4)	163 (33.7)		9,501 (95.1)	3,445 (96.5)	
Hispanic, n (%)	**Yes**	69 (1.6)	6 (1.2)	0.717	15 (1.9)	43 (1.2)	<0.001
Follow up, median [Q1,Q3]		10.0 [10.0,11.0]	5.0 [2.0,7.0]	<0.001	7.0 [4.0,10.0]	4.0 [2.0,6.0]	<0.001
BAC Grade, n (%)	**Mild**	608 (13.8)	103 (21.3)	<0.001	5,873 (58.8)	1,629 (45.6)	<0.001
	Moderate	66 (1.5)	18 (3.7)		1,631 (16.3)	674 (18.9)	
	Severe	45 (1.0)	28 (5.8)		1,537 (15.4)	1,046 (29.3)	
	Zero	3,674 (83.6)	334 (69.2)		954 (9.5)	220 (6.2)	
Smoking, n (%)	**Yes**	968 (22.0)	159 (32.9)	<0.001	3,998 (40.0)	1,771 (49.6)	<0.001
Diabetes, n (%)	**Yes**	1,496 (34.1)	302 (62.5)	<0.001	5,873 (58.8)	2,528 (70.8)	<0.001
Anti-Hypertensives, n (%)	**Yes**	559 (12.7)	163 (33.7)	<0.001	215 (2.2)	248 (6.9)	<0.001
Statins, n (%)	**Yes**	486 (11.1)	182 (37.7)	<0.001	768 (7.7)	1066 (29.9)	<0.001
Total Cholesterol, mean (SD)		191.4 (34.3)	189.8 (37.2)	0.385	188.4 (40.5)	175.8 (42.0)	<0.001
HDL, mean (SD)		59.4 (15.3)	56.0 (15.0)	<0.001	73.8 (37.5)	69.3 (36.7)	<0.001

SBP, mean (SD)	128.3 (17.6)	135.1 (20.3)	<0.001	129.6 (18.4)	132.2 (22.2)	<0.001
BMI, mean (SD)	29.1 (5.5)	30.7 (5.5)	<0.001	23.9 (15.1)	25.2 (16.9)	<0.001
eGFR, mean (SD)	83.4 (19.1)	71.5 (23.4)	<0.001	71.4 (14.0)	63.1 (17.8)	<0.001

Supplementary Table 9 Discriminative performance (C-index) of the Cox proportional hazards models

The table displays the C-index with its 95% confidence interval (CI) for each model on the Emory and Mayo test sets. All CIs were generated using the percentile method from 1,000 bootstrap iterations. Statistical significance of the difference between models was determined by a separate, **paired bootstrap analysis** (as shown in Supplementary Figure 8 Pairwise Comparison of Model Discrimination Performance), not by the overlap of these individual CIs.

	BAC modeled as Categorical or Continuous	Model	C-index (95% CI)
Internal Cohort (Emory)	Categorical	Age+BAC	0.69 (0.65 – 0.72)
		PREVENT-only	0.71 (0.68 – 0.74)
		PREVENT+BAC	0.73 (0.70 – 0.76)
	Continuous	Age+BAC	0.69 (0.66 – 0.72)
		PREVENT-only	0.74 (0.71 – 0.77)
		PREVENT+BAC	0.75 (0.72 – 0.77)
External Cohort (Mayo Clinic)	Categorical	Age+BAC	0.65 (0.63 – 0.66)
		PREVENT-only	0.62 (0.61 – 0.64)
		PREVENT+BAC	0.64 (0.63 – 0.66)
	Continuous	Age+BAC	0.66 (0.65 – 0.68)
		PREVENT-only	0.62 (0.60 – 0.64)
		PREVENT+BAC	0.64 (0.62 – 0.66)

Supplementary Table 10 Calibration performance (slope and intercept with 95% confidence intervals) of the Cox proportional hazards models on the validation and test sets for the Emory and Mayo cohorts

	BAC modeled as Categorical or Continuous	Dataset	Calibration Slope (95%CI)	Calibration Intercept (95%CI)
Emory	Categorical	Validation	1.059 (0.223 – 1.816)	-0.009 (-0.103 – 0.120)
		Test	1.041 (0.723 – 1.450)	0.021 (-0.046 – 0.064)
	Continuous	Validation	1.040 (0.531 – 1.615)	-0.002 (-0.061 – 0.056)
		Test	1.289 (1.007 – 1.570)	-0.032 (-0.060 – -0.003)
Mayo	Categorical	Validation	1.126 (0.622 – 1.788)	-0.051 (-0.242 – 0.119)
		Test	1.120 (0.831 – 1.474)	-0.048 (-0.146 – 0.048)
	Continuous	Validation	1.146 (0.713 – 1.563)	-0.076 (-0.182 – 0.056)
		Test	1.200 (1.011 – 1.449)	-0.089 (-0.153 – -0.035

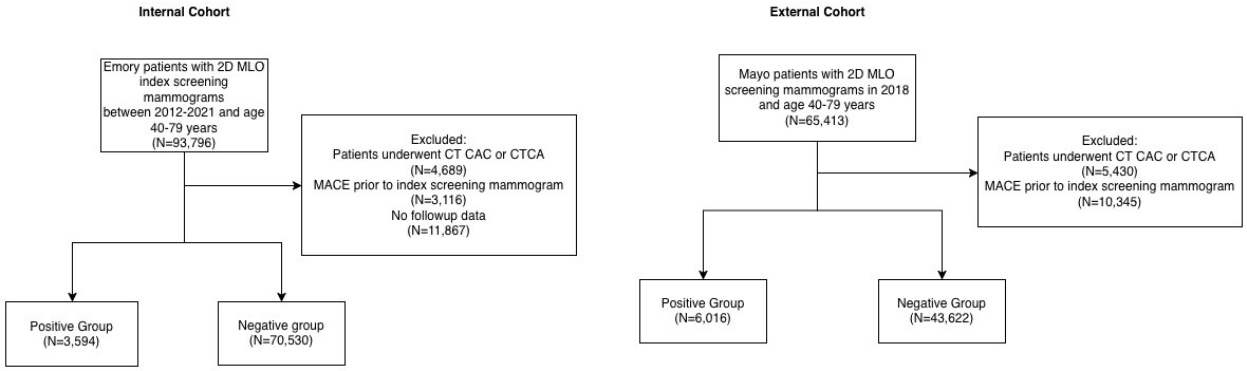
Supplementary Table 11 Distribution of Scanner Manufacturers

	Manufacturer	%
Internal Cohort	Hologic, Inc.	89.1
	GE Medical Systems	10.93
External Cohort	Hologic, Inc.	97.63
	LORAD	2.37

Supplementary Figures

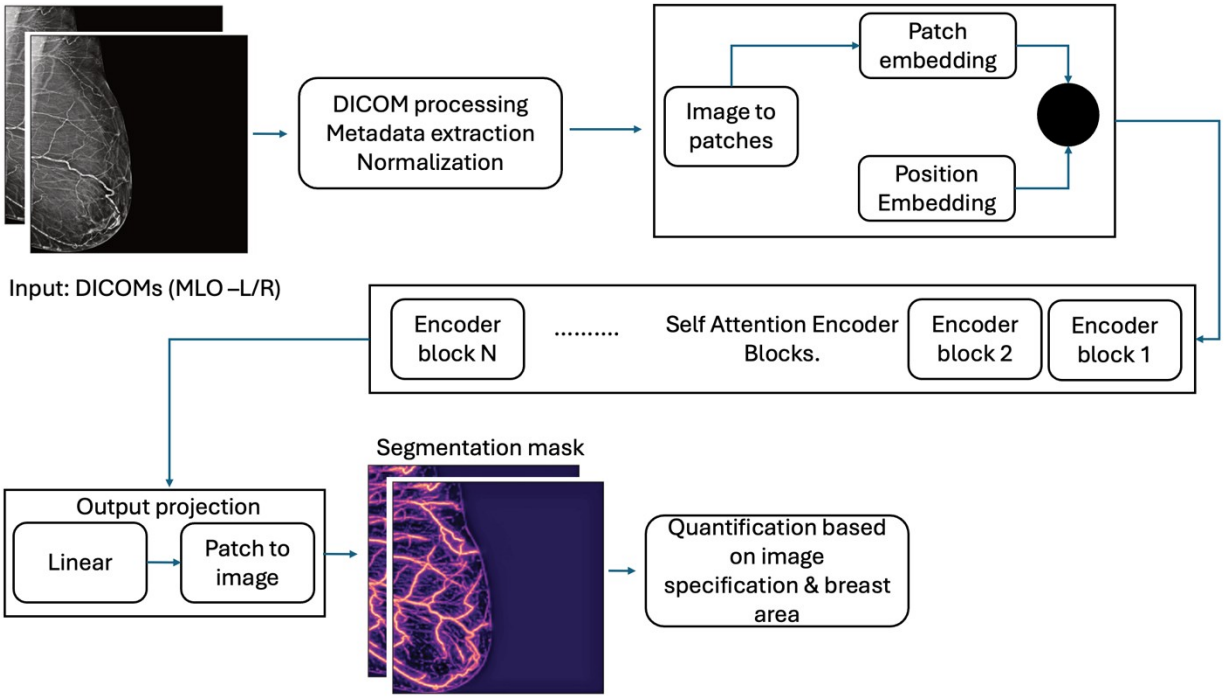
Supplementary Figure 1 Patient Flow Diagram

Patient selection flow diagram for internal and external cohort.

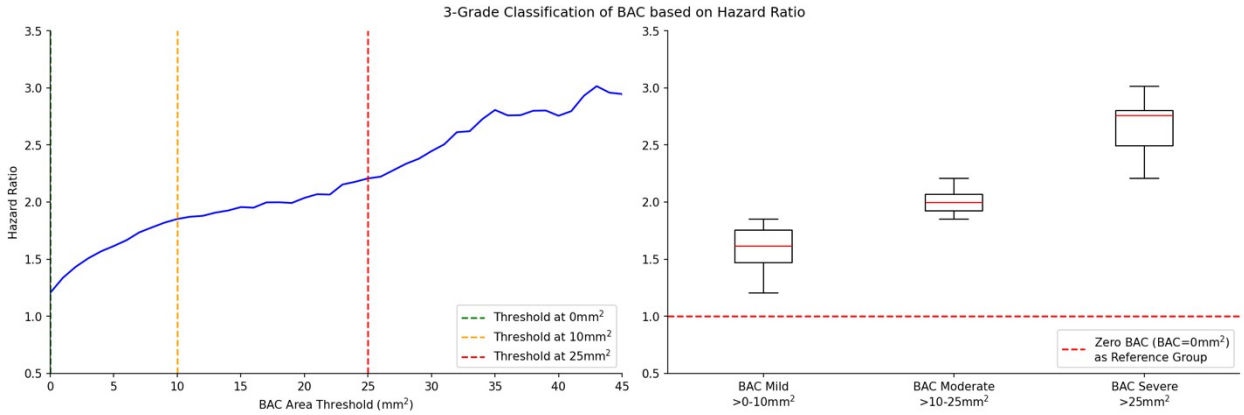


Supplementary Figure 2 Model Architecture

We developed a proprietary, transformer-based neural network architecture for the direct segmentation and quantification of Breast Arterial Calcification (BAC). The model is designed to analyze high-resolution, two-dimensional (2D) full-field digital screening mammograms (FFDM), specifically utilizing the mediolateral oblique (MLO) view.

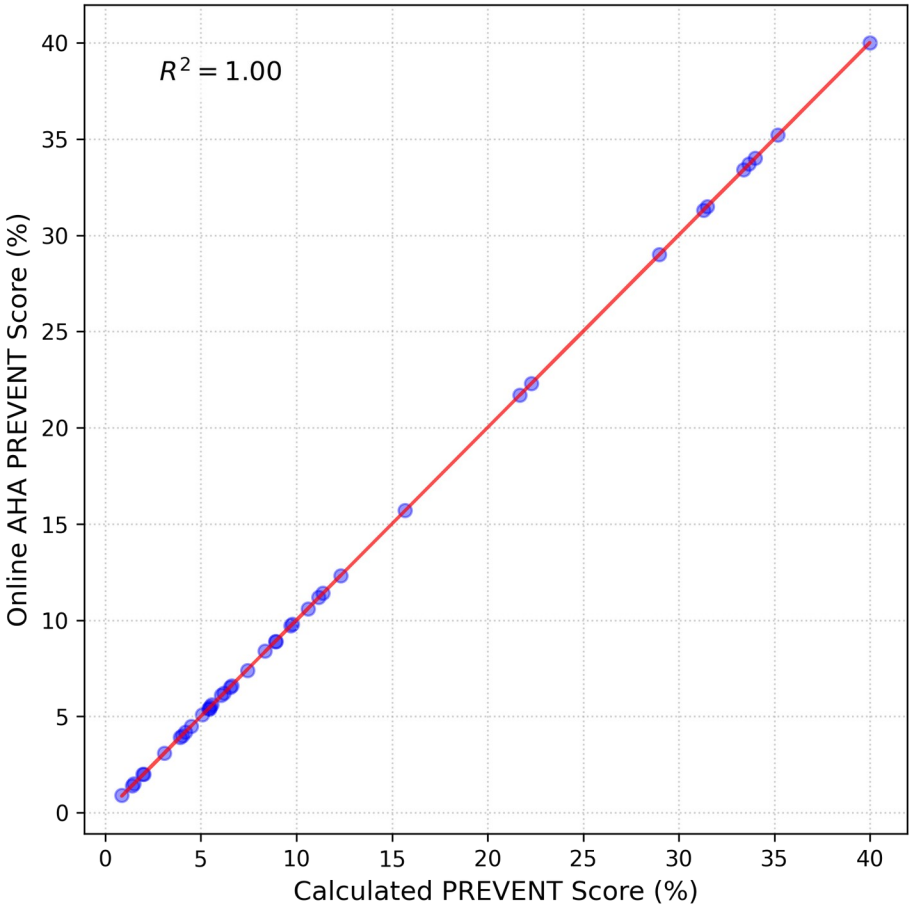


Supplementary Figure 3 BAC Severity Classification



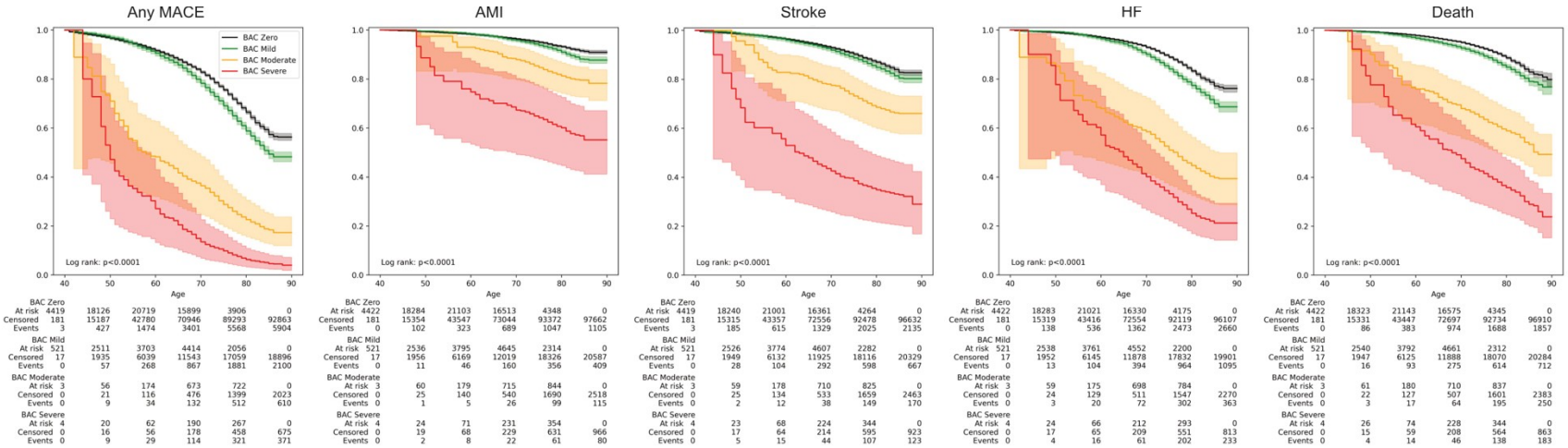
To determine optimal breast arterial calcification area (BAC) thresholds for severity classification, we performed a threshold analysis using univariate Cox proportional hazards models. We systematically evaluated different BAC cutoff values at 5mm² intervals between 0-50mm². The thresholds of 10mm² and 25mm² were selected as they maximized the hazard ratio differences between adjacent severity groups while maintaining adequate sample sizes in each group. After establishing these thresholds, we validated the groupings by comparing clinical outcomes between mild (>0-10mm²), moderate (>10-25mm²), and severe (>25mm²) groups using independent samples t-tests.

Supplementary Figure 4 Validate Calculated PREVENT Scores with Online AHA PREVENT Calculator



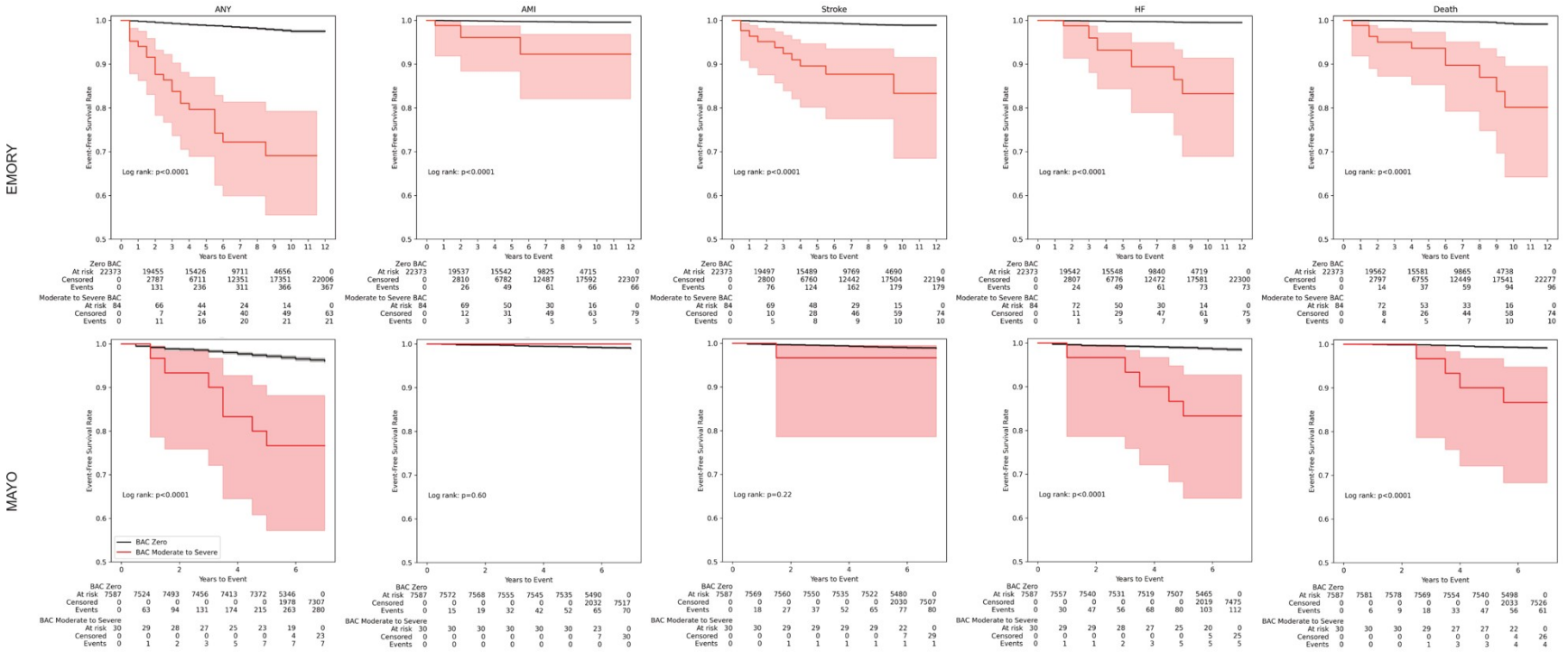
Calibration analysis was performed to assess the relationship between the PREVENT risk scores calculated by our automated script developed from the PREVENT equation and the ground truth PREVENT risk scores calculated from the online tool[2]

Supplementary Figure 5 Kaplan-Meier Curves adjusted for Age

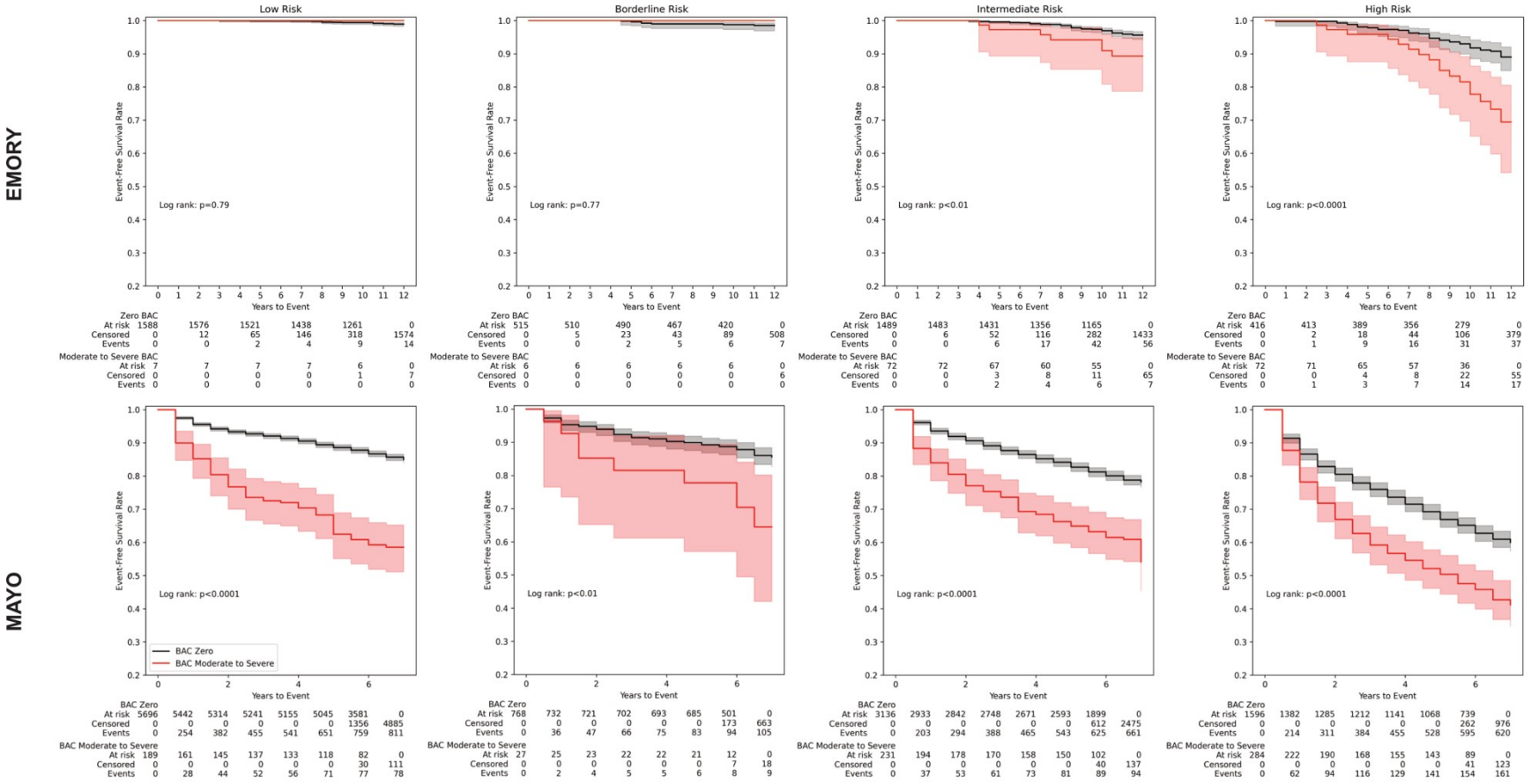


Kaplan-Meier curves of the combined Emory and Mayo cohorts across MACE events stratified by BAC severity groups with age as a timescale with left-truncation. The tables below the plots indicate the number of patients at risk for each BAC severity group at 10-year age intervals. Differences in event-free survival rates among the BAC severity groups were assessed using the log-rank test.

Supplementary Figure 6 Kaplan-Meier Curves for Internal and External Cohorts for Patients <50 Years (Zero BAC vs Moderate to Severe BAC)

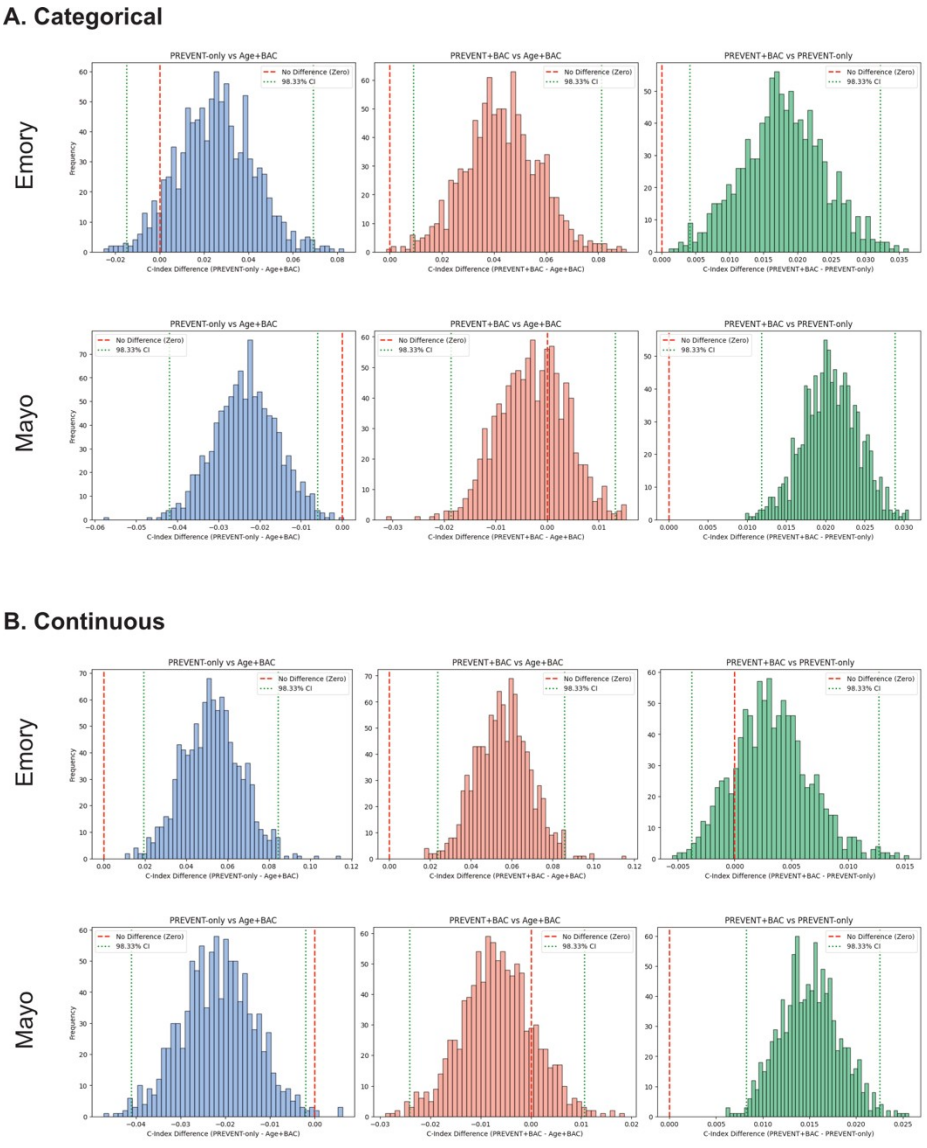


Supplementary Figure 7 Kaplan-Meier Curves for Internal and External Cohorts Stratified by PREVENT Risk Categories



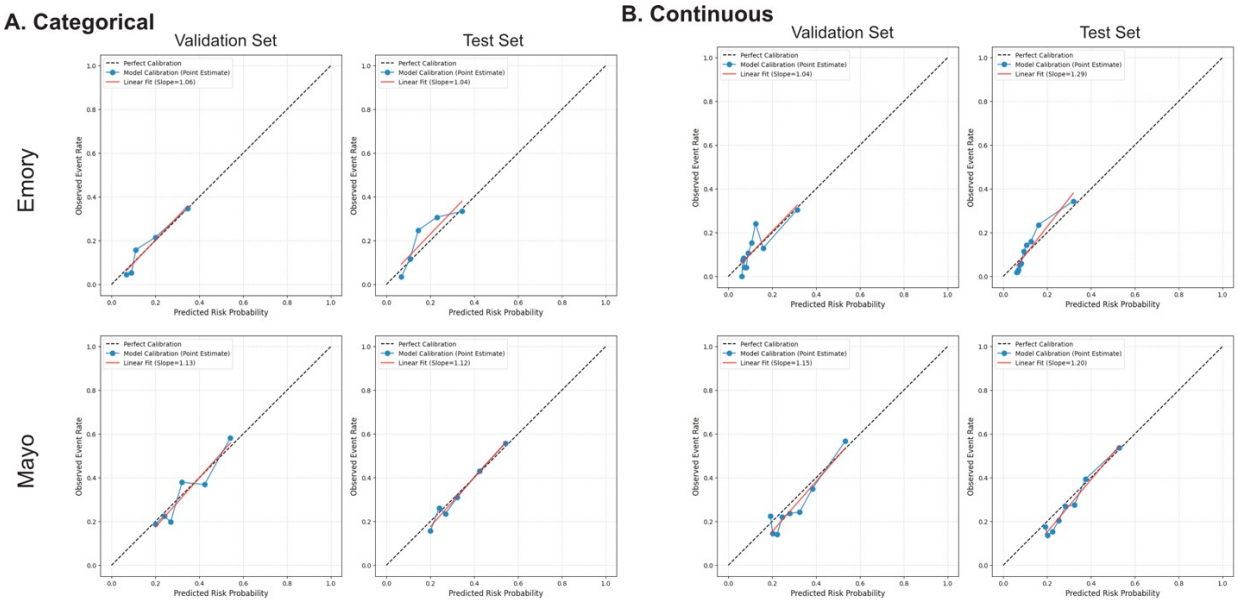
Supplementary Figure 8 Pairwise Comparison of Model Discrimination Performance

Bootstrap distributions of the difference in Harrell's C-index for pairwise model comparisons. The analysis is presented for two different modeling approaches for the breast arterial calcification (BAC) variable: A) as a categorical variable and B) as a continuous variable. Each subplot shows the distribution from 1,000 bootstrap samples, with the dashed red line at zero representing the null hypothesis of no difference. The dashed green line indicates the observed mean difference, with the corresponding Bonferroni-corrected 98.33% confidence interval also shown. The results consistently show that the PREVENT+BAC model significantly outperforms both the PREVENT-only and Age+BAC models across both cohorts and modeling approaches.



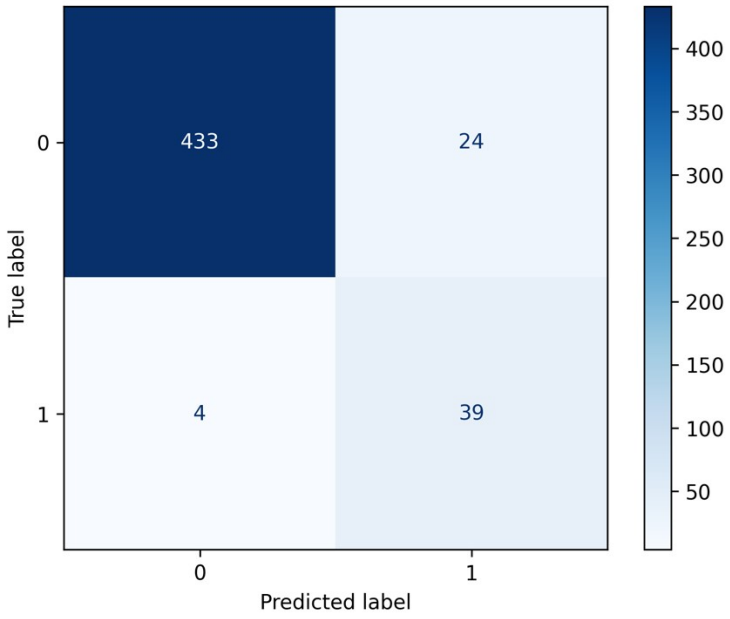
Supplementary Figure 9 Calibration of the PREVENT+BAC Model in the Emory and Mayo Clinic Cohorts

Calibration plots for the final combined model (PREVENT+BAC). The analyses are stratified by how BAC was modeled (A. Categorical vs. B. Continuous), by cohort (Emory vs. Mayo), and by dataset (Validation vs. Test). In each panel, the model-predicted risk (x-axis) is plotted against the observed event rate (y-axis) after grouping patients into quantiles based on their predicted risk. The dashed diagonal line indicates perfect calibration (slope=1.0). The model's calibration performance (blue line) and the linear fit (red line) are shown, with the overall calibration slope provided. The model demonstrates good calibration in all scenarios, with observed event rates closely aligning with the predicted probabilities.



Supplementary Figure 10 Performance of the AI model for detecting the presence of BAC on an indepdendent set of 500 exams

Summary of model performance evaluated on an independent set of 500 exams.

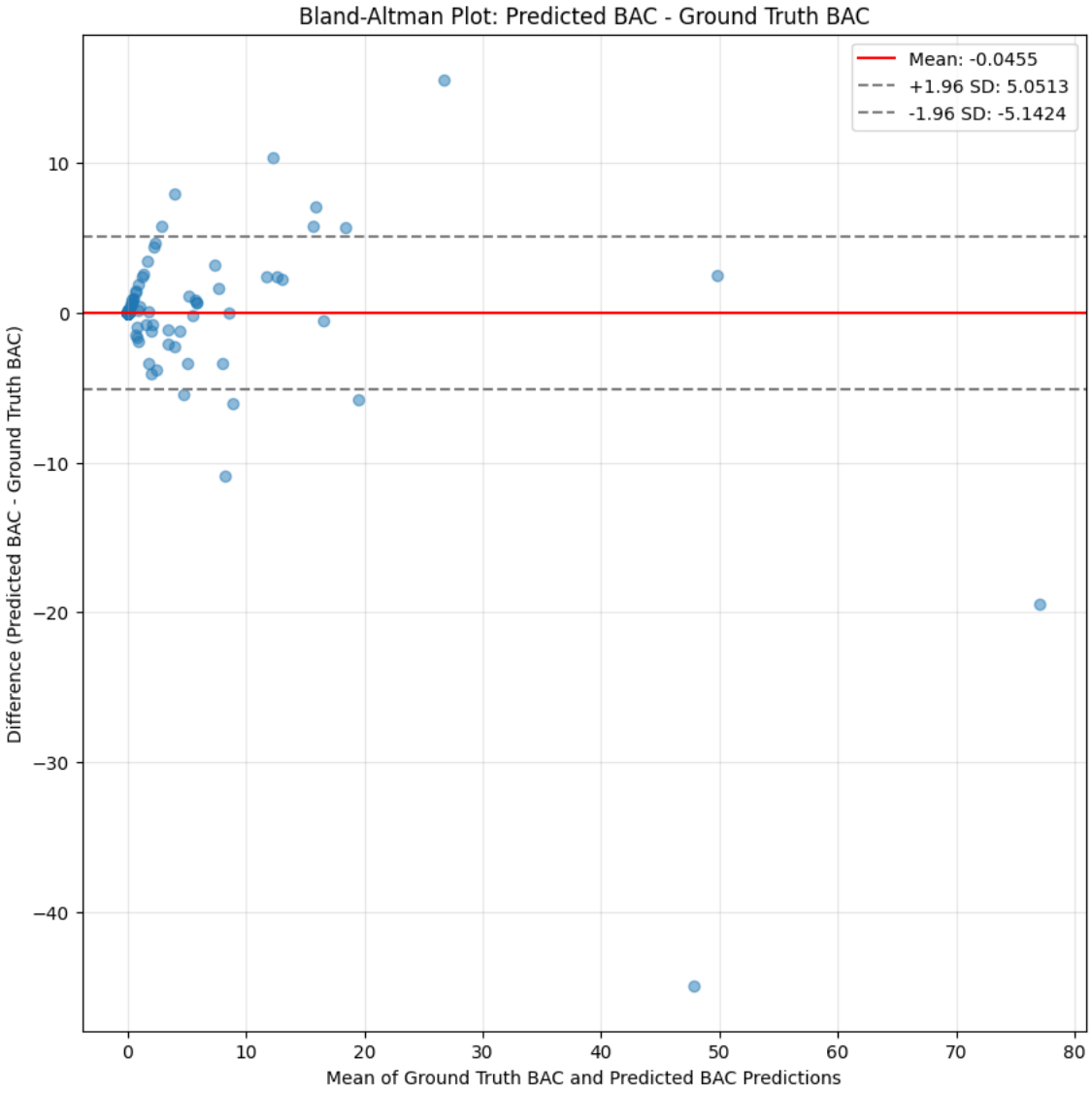


	precision	recall	f1-score	support
False	0.99	0.95	0.97	457
True	0.62	0.91	0.74	43
accuracy			0.94	500
macro avg	0.80	0.93	0.85	500
weighted avg	0.96	0.94	0.95	500

- In the True Positive group (N=39), the median predicted BAC area was 6.11mm^2 (IQR 2.18mm^2 - 15.22mm^2) and the median ground truth BAC area was 6.73mm^2 (IQR 4.4mm^2-12.17mm^2).
- In the False Negative group (N=4), the median ground truth BAC area was 1.77mm^2 (IQR 1.60mm^2 - 2.43mm^2), which would be in the Mild BAC category.
- In the False Positive group (N=24), the median predicted BAC area was 0.88mm^2 (IQR 0.15mm^2 - 2.44mm^2), which would be in the Mild BAC category.

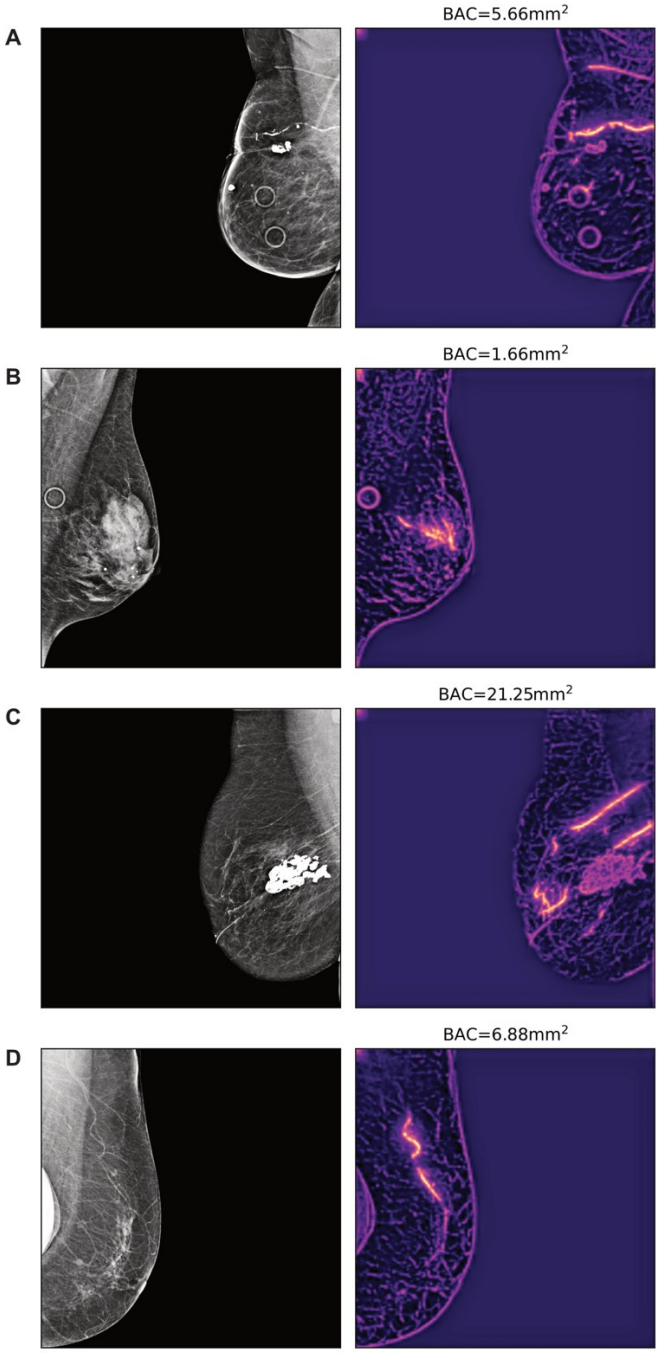
Supplementary Figure 11 Bland-Altman analysis of agreement between AI-quantified and Ground Truth BAC scores on an independent set of 500 exams

The agreement between our AI-quantified and ground truth BAC scores (n=500 exams) shows a low systematic bias with a mean bias of -0.0455 (red line), and the 95% limits of agreement between -5.14 to +5.05 (dashed lines).



Supplementary Figure 12 Examples of True Positives in the presence of non-vascular calcifications, markers, and breast implant.

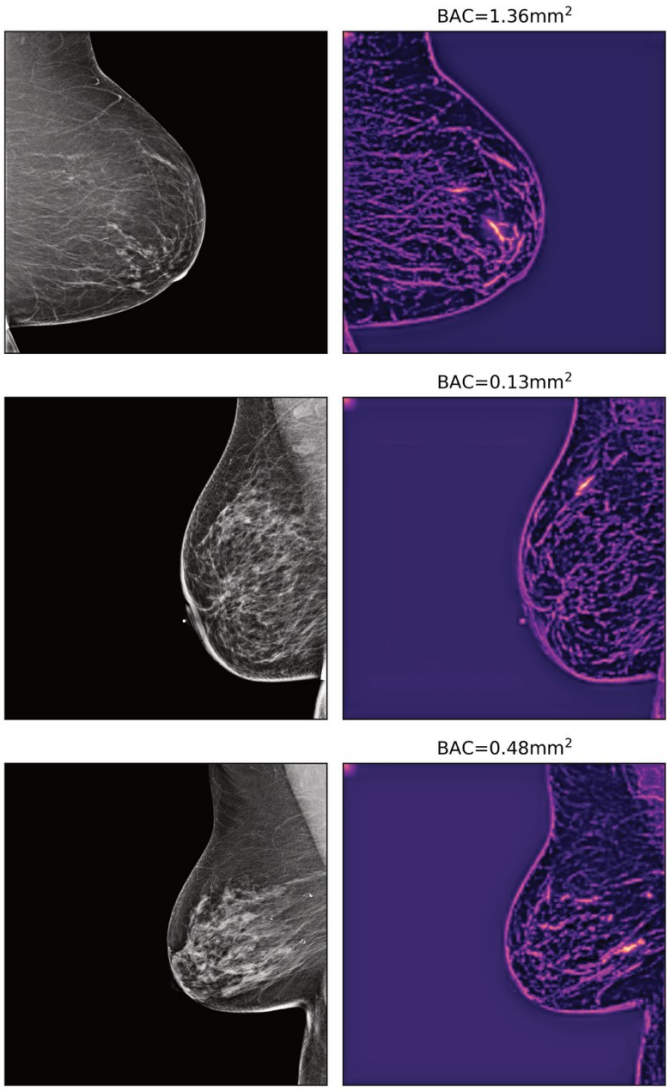
Examples of true positive BAC predictions evaluated on an independent annotated set of 500 exams. Model performance on this set is shown in Supplementary Figure 10.



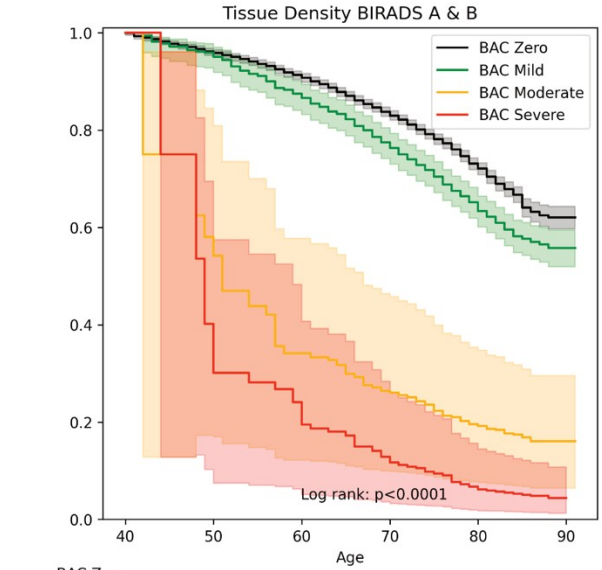
A. Resilience to mole markers; **B.** Detection of BAC on extremely dense tissue; **C.** Model ignoring a skin scar marker traversing the nipple and very dense calcifications that are in contact with BAC; **D.** BAC detection in the presence of breast implant.

Supplementary Figure 13 Examples of False Positives

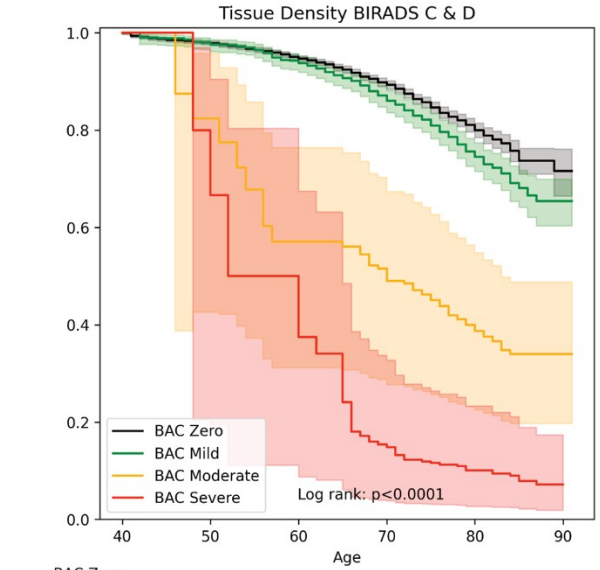
Examples of false positive BAC predictions evaluated on an independent annotated set of 500 exams. Model performance on this set is shown in Supplementary Figure 10.



Supplementary Figure 14 Kaplan-Meier analysis based on Low density vs High density breasts



BAC Zero						
At risk	1284	5031	7142	6208	1673	6
Censored	62	4234	12412	21975	28906	30451
Events	1	143	478	1115	1655	1777
BAC Mild						
At risk	96	455	873	1201	620	2
Censored	4	373	1226	2577	4034	4609
Events	0	16	75	209	389	432
BAC Moderate						
At risk	1	15	46	182	221	4
Censored	0	4	37	142	394	588
Events	0	5	14	40	104	127
BAC Severe						
At risk	2	9	24	51	78	0
Censored	0	5	23	76	153	219
Events	0	7	15	38	82	94



BAC Zero						
At risk	2349	7226	5284	3255	732	3
Censored	119	7642	17917	24636	28240	28936
Events	2	109	320	570	788	821
BAC Mild						
At risk	272	953	930	840	384	3
Censored	13	920	2377	3669	4695	5051
Events	0	17	55	131	225	250
BAC Moderate						
At risk	1	17	36	90	98	3
Censored	0	13	40	113	232	318
Events	0	2	8	19	44	53
BAC Severe						
At risk	2	5	8	28	36	0
Censored	0	9	20	38	88	118
Events	0	2	5	20	32	38

Low density breast: BIRADS A & B
High density breast: BIRADS C & D

Supplemental References

1. Bosco E, Hsueh L, McConeghy KW, Gravenstein S, Saade E. Major adverse cardiovascular event definitions used in observational analysis of administrative databases: a systematic review. *BMC Med Res Methodol*. 2021;21(1):241. doi:10.1186/s12874-021-01440-5

2. AHA. The American Heart Association PREVENT Online Calculator. professional.heart.org. 2023. Accessed May 30, 2025. https://professional.heart.org/en/guidelines-and-statements/prevent-calculator